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5 / 5 submitted

Q1. Tuple

ANSWER

A tuple cannot be changed after creation, so multiple threads can safely read the same tuple without worrying that another thread will modify it. This reduces race conditions and the need of lock when sharing the data only between threads.

Q2. SET

ANSWER

```
s={1,3,4}
```

Q3. LIST

ANSWER

```
queue = []
queue.append(10)
queue.append(20)
queue.append(30)

print("queue:", queue)
print("deleted:", queue.pop(0))
print("front:", queue[0])
print("is empty:", len(queue) == 0)

Output:
queue :[10,20,30]
deleted :10
front:20
is empty : false
```

Q4. DICTIONARY

ANSWER

A Dictionary stores data in key value pairs where each key is unique and is used to access its corresponding value.

```
ex:{"name:" "ravi" , "age:" "22"}
```

Hashing: It is a process of converting a key into hash value or index using a hash function, which helps in storing data and retrieving data quickly.

Q5. LIST

ANSWER

List Comprehension: It stores and creates the entire list in memory at once.

```
Ex: Squares=[x *x for x in range(5)]
```

Generator: Produces one value at a time, so it uses less memory.

```
Ex: Squares=(x*x for x in range(5))
```

Difference Between These is List comprehension uses more memory, while generator expression has efficient memory.